Lemma. Let G be a group.
Given $g \in G$, a map $G \to G;\ h \mapsto ghg^{-1}$ is an automorphism.
Proof. Let $g \in G$ be fixed.
1) Homomorphism.
Given $h_1, h_2 \in G$, $gh_1h_2g^{-1} = gh_1g^{-1} \cdot gh_2g^{-1}$
2) Injectivity
$gh_1g^{-1} = gh_2g^{-1} \Rightarrow h_1 = h_2$ by the cancelation.
3). Surjectivity.
Given $h \in G$, $g(g^{-1}hg)g^{-1} = h$. So, proved. □

Thm. Let G be a group. Define $Inn(G) := \{h \mapsto ghg^{-1} \mid g \in G\}$. Then,

$$Inn(G) \trianglelefteq Aut(G).$$

Proof. The inclusion is already obtained by the lemma above.
Denote $\varphi_g : G \to G;\ h \mapsto ghg^{-1}$. Since $e \in G$, $\varphi_e \in Inn(G) \neq \emptyset$.
Given $g_1, g_2 \in G$, $\varphi_{g_1} \circ (\varphi_{g_2})^{-1} = \varphi_{g_1g_2^{-1}} \in Inn(G)$, so $Inn(G) \leq Aut(G)$.

And, given $\alpha \in Aut(G)$ and $\varphi_g \in Inn(G)$,

$$(\alpha \circ \varphi_g \circ \alpha^{-1})(h) = \alpha\left(g\,\alpha^{-1}(h)\,g^{-1}\right) = \alpha(g) \cdot h \cdot \alpha(g)^{-1}, \quad \forall h \in G,$$

So $\alpha \circ \varphi_g \circ \alpha^{-1} = \varphi_{\alpha(g)} \in Inn(G)$, so proved

Def. Let G be a group. Define

$$Out(G) \overset{def}{=} Aut(G)/Inn(G).$$

Exact sequence.

Def. Let $\{G_i\}_{i=1}^{n}$ be a sequence of groups.
A sequence of Homomorphisms $\{f_i : G_i \to G_{i+1}\}_{i=1}^{n-1}$ is called exact if;

for each $1 \le i \le n-2$, $\operatorname{Im} f_i = \ker f_{i+1}$.

That is, $G_1 \xrightarrow{f_1} G_2 \xrightarrow{f_2} \cdots \to G_{n-1} \xrightarrow{f_{n-1}} G_n$.

Note: Consider an exact sequence

$$0 \xrightarrow{\text{triv.}} G_1 \xrightarrow{f} G_2 \xrightarrow{g} G_3 \xrightarrow{\text{triv.}} 0$$

Then, $\ker f = \{e_1\} \subseteq G_1$, that is, f is injective,
and, $\operatorname{Im} g = G_3$, that is, g is surjective.

Let $f : G \to H$ be a group-Homomorphism. Then,

$$0 \to \ker f \xrightarrow{f|} G \xrightarrow{\pi} G/\ker f \to 0$$